

Relative Cowedges and Coends over a Site

Motivated by Bruno Drioux's lecture series on indexed categories and relative topos theory

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Abstract

We propose a localization-sensitive universal property for ends and coends of category-valued data indexed over a Grothendieck site. The definition is made for a pseudofunctor of coefficients and specializes to the canonical stack of a sheaf topos. Its cowedges range over every generalized object above a slice site, rather than over a single fibre. We prove the representing property, uniqueness, functorial restriction to slices, invariance under indexed equivalence, and reduction to ordinary coends over the terminal site. We then give an explicit generators-and-relations coequalizer for every fixed base object. If the site category has chosen pullbacks, the coefficients have indexed coproducts, and reindexing preserves the displayed colimits, this formula is stable under base change. A pullback-deficient site shows why the extra base hypothesis cannot simply be omitted. Finally, under the precise hypothesis that the inner relative coends exist stably, we prove a Fubini theorem by identifying iterated and simultaneous relative cowedges. Relative co-Yoneda, profunctor composition, and site-presentation invariance remain open stages of the programme.

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1 Introduction

Ordinary coends package dinatural families into a single universal object. They support the calculus of Kan extensions, profunctors, density, and the co-Yoneda lemma. For a category varying over a base, however, there are two different operations that one might call a coend. One can form an ordinary coend separately in each fibre, or one can ask for a universal object that is compatible with every change of base. The first operation is familiar but does not, by itself, produce the second. The elementary example in [example 5.3](#) makes this distinction explicit.

This paper begins a coend calculus adapted to categories indexed over a Grothendieck site. Its basic move is to formulate the universal property over the whole slice site $\mathcal{C}/U$. A relative cowedge with apex X therefore has a component not only for every object in $\mathbb{D}(U)$, but for every generalized object $A \in \mathbb{D}(V)$ above every arrow $v : V \rightarrow U$. A second axiom compares these components after further pullback. The resulting universal object records change-of-base data that a fibrewise definition omits. It becomes descent-sensitive only when the indexed inputs satisfy the relevant J -descent hypotheses; the topology does not enter the two defining equations by itself.

The motivation comes from Drioux’s lecture series on indexed categories and relative topos theory [[Dri25b](#), [Dri25a](#), [Dri26a](#), [Dri26b](#)]. Those slides give an indexed Yoneda construction, formulate indexed colimits by localization to every slice, and announce relative (co-)Yoneda and indexed-(co)limit projects. They are programme-level sources, not published proofs of the claims developed here. We therefore reconstruct the definitions from their universal properties and prove independently every existence or computation statement used below.

There is important prior art. Shulman’s indexed-enriched formalism already constructs a canceling tensor product and the composition of indexed profunctors by a fibrewise coequalizer under indexed coproduct and Beck–Chevalley hypotheses [[Shu13](#), Definition 2.4, Definition 2.10, and Lemma 3.25]. Earlier, Betti and Walters developed an end calculus for categories enriched in the bicategory of spans of a base topos [[BW89](#)]. Lee develops indexed profunctors over 2-categories [[Lee23](#)]. The present definition is narrower in one direction and different in another: it is initially unenriched and tied to a fixed site, but it makes localization over every slice part of the cowedge itself. Thus the intended new problem is not merely to decorate the ordinary coend sign. It is to relate coend calculus to topology, descent, and change of site in relative topos theory.

Pitts gives an earlier and particularly close warning against claiming indexed Fubini itself as new. In Sections 1–2 of his 1985 paper, the words *category*, *functor*, and *natural transformation* mean S -indexed ones; Lemma 1.1 uses an indexed coend construction, and the proof of Theorem 2.3 explicitly invokes Fubini for iterated coends [[Pit85](#), Convention and Lemma 1.1, pp. 46–47; proof of Theorem 2.3, p. 51]. The candidate contribution here is therefore narrower: the full-slice cowedge with its base relations, its representing coequalizer, and a proof that this presentation is stable under the stated pullback hypotheses.

Main results

After fixing size and variance conventions, we:

1. define localized relative cowedges, coends, wedges, and ends;
2. identify a relative coend with a representing object for the cowedge functor;
3. construct restriction along every map of the site and the canonical base-change comparison between local coends;
4. prove invariance under a chosen indexed equivalence and reduction to the ordinary definition over the terminal site; and
5. exhibit a two-object indexed category for which fibrewise coends fail to assemble, while the localized coends are stable;
6. construct every fixed-base relative coend by an explicit coequalizer, using only the displayed coproducts, coequalizers, and left adjoints to reindexing;
7. prove stable base change when the base has chosen pullbacks, Beck–Chevalley holds, and reindexing preserves those colimits; and
8. prove a Fubini theorem whenever the inner relative coends exist and are stable under base change.

All these statements are proved in this paper. We do *not* yet claim a relative co-Yoneda lemma, a localization-sensitive profunctor bicategory, or invariance under a change of site presentation. Those are separated into explicit problems in [section 10](#).

2 The ambient indexed data

2.1 Sites and relative-topos conventions

Fix Grothendieck universes $\mathcal{U} \in \mathcal{W}$ and a $\mathcal{U}$ -small site $(\mathcal{C}, J)$. In this paper **Cat** denotes $\mathcal{U}$ -small categories, while **CAT** allows $\mathcal{W}$ -small categories with $\mathcal{U}$ -small hom-sets. Thus the indexing collections and the sets of cowedge components below are $\mathcal{U}$ -small; the coefficient categories may have a $\mathcal{W}$ -small class of objects. Write

$$\mathcal{E} = \mathbf{Sh}(\mathcal{C}, J), \quad \ell_J = a_{JY} : \mathcal{C} \longrightarrow \mathcal{E}$$

for the associated sheaf topos and the sheafified Yoneda functor. The canonical stack attached to this presentation is the pseudofunctor

$$\mathbb{S}_{(\mathcal{C}, J)} : \mathcal{C}^{\text{op}} \longrightarrow \mathbf{CAT}, \quad U \longmapsto \mathcal{E}/\ell_J(U),$$

whose reindexing functors are pullbacks. Caramello and Zafra prove that this is a J -stack [CZ21, Definition 2.6.4 and Theorem 2.6.8]. It will be the principal coefficient system for applications, although the definition below works for a general indexed category of coefficients.

In the terminology of relative topos theory, a relative topos over $\mathcal{E}$ is a geometric morphism $p : \mathcal{F} \rightarrow \mathcal{E}$ [CZ21, Definition 8.2.2]. We keep this distinct from an indexed category $\mathbb{D}$: the latter is input from which relative categorical constructions are formed, not by itself a relative topos.

The smallness assumption is a working convention. A small-generated site is locally small and has a small J -dense subcategory [CZ21, p. 4]. After choosing such a subcategory one may apply

the same formulas to the resulting small presentation; independence of that choice is a separate comparison theorem and is not assumed below. This distinction is essential: a formula on a presentation is not automatically an invariant of the base topos.

Remark 2.1 (Where the topology enters). The equations in [definition 3.1](#) use the underlying category $\mathcal{C}$, not the covering families of J directly. The topology enters when $\mathbb{D}$ and $\mathbb{V}$ carry J -descent, when H respects that descent, or through the choice $\mathbb{V} = \mathbb{S}_{(\mathcal{C}, J)}$. Without such hypotheses the construction is localization-sensitive but not yet topology-invariant.

2.2 Indexed categories, opposites, and coefficients

Let

$$\mathbb{D} : \mathcal{C}^{\text{op}} \longrightarrow \mathbf{Cat}, \quad \mathbb{V} : \mathcal{C}^{\text{op}} \longrightarrow \mathbf{CAT}$$

be pseudofunctors. We assume that each $\mathbb{D}(U)$ is small and that each $\mathbb{V}(U)$ is locally small. For an arrow $u : U' \rightarrow U$, both reindexing functors are denoted u^* ; the ambient pseudofunctor will always be clear. The coherence isomorphism is written

$$\alpha_{v,u,X} : v^* u^* X \xrightarrow{\cong} (uv)^* X.$$

We suppress unit constraints when no ambiguity can result.

Let $\mathbb{D}^\vee$ denote the fibrewise opposite indexed category: its fibre at U is $\mathbb{D}(U)^{\text{op}}$, with the induced pseudofunctorial coherence. This is the indexed opposite used in relative category theory [[CZ21](#), Definition 2.1.4]. Fix an indexed functor

$$H : \mathbb{D}^\vee \times_{\mathcal{C}} \mathbb{D} \longrightarrow \mathbb{V}.$$

Thus each fibre has a functor

$$H_U : \mathbb{D}(U)^{\text{op}} \times \mathbb{D}(U) \longrightarrow \mathbb{V}(U),$$

and every $u : U' \rightarrow U$ has an invertible comparison

$$\chi_{u,A,B}^H : u^* H_U(A, B) \xrightarrow{\cong} H_{U'}(u^* A, u^* B), \tag{1}$$

natural in A, B and coherent under composition. Requiring χ^H to be invertible is the point at which H is genuinely indexed rather than merely laxly indexed.

For $U \in \mathcal{C}$, the localization of $\mathbb{D}$ at U is

$$\mathbb{D}/U : (\mathcal{C}/U)^{\text{op}} \longrightarrow \mathbf{Cat}, \quad (v : V \rightarrow U) \longmapsto \mathbb{D}(V).$$

If $X \in \mathbb{V}(U)$, its cartesian constant section over $\mathcal{C}/U$ has value $v^* X$ at $v : V \rightarrow U$. These two constructions supply the domain and apex of the relative cowedges below.

3 Localized cowedges and relative coends

Recall that for an ordinary functor $K : \mathcal{A}^{\text{op}} \times \mathcal{A} \rightarrow \mathcal{B}$, a cowedge to X is a dinatural family $K(A, A) \rightarrow X$; its initial cowedge is the coend [[Lor23](#), Definition 1.1.4 and Definition 1.1.6]. We add a base coherence condition to this familiar definition.

Definition 3.1 (Localized relative cowedge). Fix $U \in \mathcal{C}$ and $X \in \mathbb{V}(U)$. A *relative cowedge from H to X over U* is a family

$$\omega_{v,A} : H_V(A, A) \longrightarrow v^* X \tag{2}$$

for every arrow $v : V \rightarrow U$ and every $A \in \mathbb{D}(V)$, satisfying:

(C1) *fibrewise dinaturality*: for each $f : A \rightarrow B$ in $\mathbb{D}(V)$,

$$\omega_{v,A} H_V(f^{\text{op}}, 1_A) = \omega_{v,B} H_V(1_B, f) : H_V(B, A) \longrightarrow v^* X; \quad (3)$$

(C2) *base coherence*: for every $w : W \rightarrow V$,

$$\omega_{vw, w^* A} \chi_{w, A, A}^H = \alpha_{w, v, X} w^*(\omega_{v, A}) : w^* H_V(A, A) \longrightarrow (vw)^* X. \quad (4)$$

The set of such families is denoted $\text{Cow}_U^{\text{rel}}(H; X)$.

The definition ranges over the Grothendieck construction of $\mathbb{D}/U$, but writing out equations (3) and (4) avoids choosing a cleavage for that construction. The two equations have different jobs: dinaturality moves within one fibre, whereas base coherence moves between fibres.

A morphism $(X, \omega) \rightarrow (Y, \tau)$ is a morphism $m : X \rightarrow Y$ in $\mathbb{V}(U)$ such that

$$v^* m \omega_{v, A} = \tau_{v, A}$$

for all v, A . This defines the category $\mathbf{Cow}_U(H)$. Equivalently, $\mathbf{Cow}_U(H)$ is the category of elements of the covariant functor

$$\text{Cow}_U^{\text{rel}}(H; -) : \mathbb{V}(U) \longrightarrow \mathbf{Set}.$$

Definition 3.2 (Relative coend). A *relative coend* of H over U is an initial object (L_U, λ^U) of $\mathbf{Cow}_U(H)$. When it exists, we write

$$L_U = \int_{\text{rel}, U}^{A \in \mathbb{D}} H(A, A).$$

The notation records that the cowedge is localized over $\mathcal{C}/U$; it is not an assertion that an ordinary fibrewise coend has already been computed.

Proposition 3.3 (Representing property and uniqueness). *For $L_U \in \mathbb{V}(U)$, the following are equivalent.*

1. *There is a relative cowedge λ^U making (L_U, λ^U) initial in $\mathbf{Cow}_U(H)$.*
2. *There is a natural isomorphism*

$$\mathbb{V}(U)(L_U, X) \cong \text{Cow}_U^{\text{rel}}(H; X) \quad (5)$$

sending 1_{L_U} to λ^U .

Consequently, a relative coend is unique up to a unique isomorphism compatible with its universal cowedge.

Proof. Given an initial cowedge, send $m : L_U \rightarrow X$ to the family $(v^* m \lambda_{v, A}^U)_{v, A}$. Initiality gives the inverse and its uniqueness, while composition in $\mathbb{V}(U)$ gives naturality in X . Conversely, the image of 1_{L_U} under equation (5) is a universal element, hence an initial cowedge. The final assertion is the standard uniqueness of representing objects, including compatibility with the distinguished universal element. $\square$

Definition 3.4 (Relative wedge and end). For an indexed functor $K : \mathbb{D}^\vee \times_{\mathcal{C}} \mathbb{D} \rightarrow \mathbb{V}$, a relative wedge from $X \in \mathbb{V}(U)$ to K is a family

$$\kappa_{v,A} : v^* X \longrightarrow K_V(A, A)$$

for $v : V \rightarrow U$ and $A \in \mathbb{D}(V)$, subject to two equations. For $f : A \rightarrow B$ in $\mathbb{D}(V)$,

$$K_V(1_A, f) \kappa_{v,A} = K_V(f^{\text{op}}, 1_B) \kappa_{v,B} : v^* X \longrightarrow K_V(A, B), \quad (6)$$

and for $w : W \rightarrow V$,

$$\kappa_{vw, w^* A} \alpha_{w,v,X} = \chi_{w,A,A}^K w^* \kappa_{v,A} : w^* v^* X \longrightarrow K_W(w^* A, w^* A). \quad (7)$$

A morphism $(X, \kappa) \rightarrow (Y, \tau)$ is a map $m : X \rightarrow Y$ with $\kappa_{v,A} = \tau_{v,A} v^* m$ for every v, A . These objects and morphisms form $\text{Wedge}_U(K)$. A *relative end* is a terminal object of $\text{Wedge}_U(K)$, written

$$\int_{A \in \mathbb{D}}^{\text{rel}, U} K(A, A).$$

No completeness of the fibrewise opposite coefficient system is assumed.

4 Restriction and stable base change

The use of all generalized objects has an immediate payoff: cowedges restrict without any existence assumption.

Proposition 4.1 (Restriction to a slice). *Every arrow $u : U' \rightarrow U$ induces a functor*

$$\rho_u : \mathbf{Cow}_U(H) \longrightarrow \mathbf{Cow}_{U'}(H).$$

It sends an apex X to $u^ X$. Up to the pseudofunctorial constraints, the component of $\rho_u \omega$ at $v : V \rightarrow U'$ and $A \in \mathbb{D}(V)$ is $\omega_{uv,A}$. These functors are pseudofunctorial: for $U'' \xrightarrow{v} U' \xrightarrow{u} U$, the compositor*

$$\gamma_{v,u} : \rho_v \rho_u \Longrightarrow \rho_{uv}$$

has apex component $\alpha_{v,u,X} : v^ u^* X \rightarrow (uv)^* X$, and the unitor is the unit constraint of $\mathbb{V}$.*

Proof. Define

$$(\rho_u \omega)_{v,A} : H_V(A, A) \xrightarrow{\omega_{uv,A}} (uv)^* X \xrightarrow{\alpha_{v,u,X}^{-1}} v^* u^* X.$$

The dinaturality equation is exactly that for $\omega_{uv,-}$. Base coherence follows by pasting the base-coherence square for ω with the associativity coherence for α . On morphisms put $\rho_u(m) = u^* m$. For the compositor, the defining equality for a morphism of cowedges is the associativity pentagon for α , applied to $\omega_{uvz,A}$; hence its apex is exactly $\alpha_{v,u,X}$. The unitor check is the triangle axiom. The pentagon and triangle for γ are therefore those of the pseudofunctor $\mathbb{V}$, component by component. $\square$

Suppose relative coends (L_U, λ^U) have been chosen for all $U \in \mathcal{C}$. The restricted cowedge $\rho_u \lambda^U$ has apex $u^* L_U$. Initiality at U' therefore supplies a unique comparison

$$b_u : L_{U'} \longrightarrow u^* L_U \quad (8)$$

such that $b_u \lambda^{U'} = \rho_u \lambda^U$, with the structural isomorphisms understood.

Definition 4.2 (Stable relative coends). A chosen family of relative coends is *stable under base change* if every canonical map b_u in [equation \(8\)](#) is an isomorphism.

Theorem 4.3 (Coherent assembly). *The canonical maps b_u satisfy the identity and composition laws up to the coherence of $\mathbb{V}$. If the relative coends are stable, the inverses*

$$b_u^{-1} : u^* L_U \xrightarrow{\cong} L_{U'}$$

make $(L_U)_{U \in \mathcal{C}}$ a cartesian pseudonatural object of $\mathbb{V}$. This structure is uniquely determined by the universal cowedges.

Proof. For an identity arrow, both the unit comparison and b_{1_U} are morphisms from the initial cowedge to its unit restriction; they agree by uniqueness. For composable arrows $U'' \xrightarrow{v} U' \xrightarrow{u} U$, the two morphisms from the initial cowedge over U'' to the (uv) -restriction of λ^U are equal. On apexes this says

$$b_{uv} = \alpha_{v,u,L_U} v^* b_u b_v. \quad (9)$$

If all b_u are invertible, put $\phi_u = b_u^{-1} : u^* L_U \rightarrow L_{U'}$. Equation (9) is equivalent to

$$\phi_v v^* \phi_u = \phi_{uv} \alpha_{v,u,L_U}, \quad (10)$$

and the identity argument gives its unit law. Thus $U \mapsto L_U$, with the invertible cartesian transition maps ϕ_u , is a cartesian section of the Grothendieck fibration of $\mathbb{V}$, equivalently a pseudonatural transformation from the terminal pseudofunctor to $\mathbb{V}$. Any other compatible transitions would invert morphisms with the same universal property and hence coincide with them. $\square$

Remark 4.4. Existence and stability are logically separate. The definition gives b_u only after relative coends exist on both slices; it does not force that map to be invertible. A useful existence theorem must therefore either prove stability or state it as an additional hypothesis.

5 First comparison theorems

Proposition 5.1 (Indexed-equivalence invariance). *Let $F : \mathbb{D} \rightleftarrows \mathbb{D}' : G$ be a chosen pseudonatural adjoint equivalence of indexed categories, with invertible indexed unit $\eta : 1_{\mathbb{D}} \Rightarrow GF$ and counit $\varepsilon : FG \Rightarrow 1_{\mathbb{D}'}$. Let $H' : (\mathbb{D}')^\vee \times_{\mathcal{C}} \mathbb{D}' \rightarrow \mathbb{V}$, and suppose an indexed natural isomorphism*

$$\theta : H \xrightarrow{\cong} H' (F^\vee \times F)$$

has been fixed. Then for every U there is an equivalence

$$\mathbf{Cow}_U(H) \simeq \mathbf{Cow}_U(H')$$

that is the identity on apex objects. In particular, one side has a relative coend if and only if the other does, and the resulting apex objects are canonically isomorphic relative to the chosen equivalence data.

Proof. For an H' -cowedge τ with apex X , define $T\tau$ by

$$H_V(A, A) \xrightarrow{\theta_{V,A,A}} H'_V(F_V A, F_V A) \xrightarrow{\tau_{v,F_V A}} v^* X. \quad (11)$$

Naturality and indexed coherence of θ give (C1) and (C2). Conversely, for an H -cowedge ω , define $S\omega$ at $A' \in \mathbb{D}'(V)$ as

$$\begin{aligned} H'_V(A', A') &\xrightarrow{H'_V(\varepsilon_{V,A'}^{\text{op}}, \varepsilon_{V,A'}^{-1})} H'_V(F_V G_V A', F_V G_V A') \\ &\xrightarrow{\theta_{V,G_V A', G_V A'}^{-1}} H_V(G_V A', G_V A') \xrightarrow{\omega_{v,G_V A'}} v^* X. \end{aligned} \quad (12)$$

Here the first arrow is well typed because the first variable is contravariant. Naturality of ε proves (C1), while the fact that ε and θ are indexed modifications proves (C2). Both S and T leave apexes and apex morphisms unchanged.

At an object $A \in \mathbb{D}(V)$, the component of $TS\omega$ is obtained from $\omega_{v, G_V F_V A}$ along $\eta_{V, A} : A \rightarrow G_V F_V A$. Dinaturality of ω , followed by the triangle identity $\varepsilon_{FA} F \eta_A = 1_{FA}$, identifies it with $\omega_{v, A}$. The same calculation for $A' \in \mathbb{D}'(V)$, now using $G \varepsilon_{A'} \eta_{GA'} = 1_{GA'}$, identifies $ST\tau$ with τ . These componentwise identifications are indexed by coherence of $\eta, \varepsilon, \theta$, and are the identity on apexes. Thus S and T are inverse equivalences of cowedge categories. Initial objects correspond, and their apexes are uniquely isomorphic by [proposition 3.3](#). $\square$

The word *chosen* in [proposition 5.1](#) is deliberate. An unstructured statement that the fibres merely happen to be equivalent does not provide the coherence needed to transport [equation \(4\)](#).

Proposition 5.2 (Terminal-site reduction). *Let $\mathcal{C} = \mathbf{1}$ with its trivial (coarsest) Grothendieck topology. If $\mathbb{D}(\ast) = \mathcal{A}$, $\mathbb{V}(\ast) = \mathcal{B}$, and $H_\ast : \mathcal{A}^{\text{op}} \times \mathcal{A} \rightarrow \mathcal{B}$, then the relative-cowedge category over $\ast$ is canonically isomorphic, through the pseudofunctor unit constraints, to the ordinary cowedge category for $H_\ast$. Consequently,*

$$\int_{\text{rel}, \ast}^{A \in \mathbb{D}} H(A, A) \cong \int^{A \in \mathcal{A}} H_\ast(A, A)$$

whenever either side exists.

Proof. The slice $\mathbf{1}/\ast$ has one object and one arrow. Thus [equation \(2\)](#) is an ordinary family indexed by $A \in \mathcal{A}$, [equation \(3\)](#) is ordinary dinaturality, and [equation \(4\)](#) reduces to the unit coherence. Normalizing the pseudofunctors makes the two definitions literal; without normalization, the unit constraints give the asserted canonical isomorphism. $\square$

Example 5.3 (Fibrewise coends do not automatically assemble). Let $\mathcal{C}$ be the category $0 \xrightarrow{u} 1$, equipped with the trivial topology, and let $\mathbb{V}$ be the constant indexed category **Set**. Define

$$\mathbb{D}(1) = \emptyset, \quad \mathbb{D}(0) = \mathbf{1},$$

with the unique reindexing functor $u^\ast : \emptyset \rightarrow \mathbf{1}$. Let H_1 be the unique functor with empty domain and let $H_0(\ast, \ast) = \{\ast\}$.

The ordinary fibrewise coend is $\emptyset$ over 1 and a singleton over 0. Since $u^\ast$ on **Set** is the identity, these two objects cannot be the values of a cartesian indexed object: there is not even a map $\{\ast\} \rightarrow \emptyset$ in the direction of [equation \(8\)](#).

The localized relative coends behave differently. Over 0, a relative cowedge to X is one map $\{\ast\} \rightarrow X$, hence a pointed set. Over 1, the fibre $\mathbb{D}(1)$ contributes no component, but the generalized object $\ast \in \mathbb{D}(0)$ above $u : 0 \rightarrow 1$ again contributes one point of X . Thus $\mathbf{COW}_0(H)$ and $\mathbf{COW}_1(H)$ are both the category of pointed sets. Their initial objects are singletons, and the base-change map is the identity. In this example the localization-sensitive definition restores stable assembly precisely by remembering the object visible after pullback.

6 Interface with relative topos theory and prior calculus

6.1 The canonical stack as coefficients

Taking $\mathbb{V} = \mathbb{S}_{(\mathcal{C}, J)}$ places the apex of the relative coend over U in the slice topos $\mathcal{E}/\ell_J(U)$. Pullback supplies the reindexing used in [equation \(4\)](#), so a chosen stable family of relative coends

determines a cartesian section of the Grothendieck fibration of the canonical stack. This is a direct fixed-site link to relative topos theory, not yet an intrinsic relative-topos construction. A further identification of such objects with a construction intrinsic to an arbitrary geometric morphism $\mathcal{F} \rightarrow \mathcal{E}$ requires a descent theorem and is not used here.

Drieux’s indexed hom construction is the motivating special case. In the notation of the slides, its value above U has generalized components

$$[v : V \rightarrow U] \mapsto \mathbb{D}(V)(v^*Y, v^*X), \quad (13)$$

followed, when required, by sheafification. The 2026 slides then formulate an indexed colimit through universal properties on every localization and state an indexed co-Yoneda computation [Dri26a, logical slides 13–20] [Dri26b, logical slides 13–22]. Formula [equation \(13\)](#) supports the shape of [definition 3.1](#); the slides do not supply a published proof of a general relative-coend theorem, so none is inferred from them.

6.2 Relation to Shulman’s indexed profunctors

For an S -indexed symmetric monoidal category with indexed coproducts, Shulman defines a canceling tensor product. Under Beck–Chevalley, tensor preservation, and fibrewise coequalizer hypotheses, the composite of indexed profunctors is computed by a coend-shaped coequalizer [Shu13, Definition 2.4, Definition 2.10, and Lemma 3.25, pp. 7–8 and 21–22]. This is an established indexed coend calculus and must be treated as the principal comparison theory, not as an anticipated application of the present paper.

The two constructions have different inputs. Shulman’s formula begins with an indexed monoidal base and packages fibrewise coproducts together with change-of-base adjoints. [Definition 3.1](#) instead begins with an indexed functor H and enlarges each cowedge over U to every object of $\mathcal{C}/U$. A substantive comparison theorem should identify hypotheses under which Shulman’s canceling tensor represents the localized cowedge functor. Until that theorem is proved, the two constructions are related models rather than interchangeable notation.

Betti–Walters is an equally important novelty boundary: the title and published abstract of their paper identify an end calculus over a base topos in the span-enriched setting [BW89]. The construction here is instead written on a site presentation and builds generalized objects in every slice into the cowedge. Since the present source audit did not recover a full-text comparison at theorem level, no equivalence between the two formalisms is asserted without a separate comparison theorem.

6.3 Presentation boundary

An internal category in $\mathcal{E}$, a stack on $(\mathcal{C}, J)$, and a pseudofunctor on a chosen site are connected by comparison results, but they are not definitionally identical. Likewise, equivalent sites can present the same topos without giving literally equal indexing categories. The definition in [definition 3.1](#) is therefore a fixed-presentation construction. Its invariance under indexed equivalence ([proposition 5.1](#)) is a first step, but it is not yet a Morita or change-of-site invariance theorem.

7 Existence by generators and relations

Drieux’s 2026 slides propose computing indexed colimits from cocompleteness, change-of-base adjoints, and Beck–Chevalley [Dri26b, logical slides 17–18]. The slides do not give a published proof of the localized coend defined in [definition 3.1](#). We therefore begin with a fixed base object

and derive the representing object directly. The result is deliberately stated with only the colimits actually used.

7.1 A mate calculation

Assume for the moment that every reindexing functor $v^* : \mathbb{V}(U) \rightarrow \mathbb{V}(V)$ has a chosen left adjoint $v_! \dashv v^*$, with unit η^v and counit ε^v . For composable arrows

$$W \xrightarrow{w} V \xrightarrow{v} U,$$

both $(vw)_!$ and $v_!w_!$ are left adjoint, up to the constraint $\alpha_{w,v} : w^*v^* \cong (vw)^*$, to the same functor. Uniqueness of left adjoints gives a canonical natural isomorphism

$$c_{v,w} : (vw)_! \xrightarrow{\cong} v_!w_!.$$

Its direction and normalization are fixed by requiring precomposition with $c_{v,w}$ to be the natural chain

$$\begin{aligned} \mathbb{V}(U)(v_!w_!Z, X) &\cong \mathbb{V}(W)(Z, w^*v^*X) \\ &\cong \mathbb{V}(W)(Z, (vw)^*X) \cong \mathbb{V}(U)((vw)_!Z, X). \end{aligned}$$

The uniqueness part of the adjunction correspondence makes the $c_{v,w}$ coherent for triples; this is the usual composition constraint obtained by transporting composite adjunctions. For $Z \in \mathbb{V}(V)$, put

$$\beta_{v,w,Z} : (vw)_!w^*Z \xrightarrow{c_{v,w}} v_!w_!w^*Z \xrightarrow{v_!\varepsilon_Z^w} v_!Z. \quad (14)$$

Lemma 7.1 (The base-coherence mate). *Let $h : Z \rightarrow v^*X$, and let $\bar{h} : v_!Z \rightarrow X$ be its adjunct. Under $(vw)_! \dashv (vw)^*$, the adjunct of*

$$w^*Z \xrightarrow{w^*h} w^*v^*X \xrightarrow{\alpha_{w,v,X}} (vw)^*X$$

is the composite

$$(vw)_!w^*Z \xrightarrow{\beta_{v,w,Z}} v_!Z \xrightarrow{\bar{h}} X.$$

Proof. The adjunct of h is $\bar{h} = \varepsilon_X^v v_!h$. Hence the second displayed composite is

$$(vw)_!w^*Z \xrightarrow{c_{v,w}} v_!w_!w^*Z \xrightarrow{v_!\varepsilon_Z^w} v_!Z \xrightarrow{v_!h} v_!v^*X \xrightarrow{\varepsilon_X^v} X.$$

Naturality of ε^w moves $w_!w^*h$ across ε_Z^w . The result is the counit formula for the composite adjunction $v_!w_! \dashv w^*v^*$. Transporting that composite adjunction along $c_{v,w}$ and $\alpha_{w,v}$ gives the counit formula for $(vw)_! \dashv (vw)^*$. Taking its adjunct therefore yields $\alpha_{w,v,X}w^*h$, as asserted. This is precisely the standard calculus of mates; the displayed computation fixes the direction of every constraint used later. $\square$

7.2 The coequalizer presentation

Fix $U \in \mathcal{C}$. Since $\mathcal{C}$ and every $\mathbb{D}(V)$ are small, the following indexing collections are sets. Define the object of generators

$$P_U = \coprod_{\substack{v: V \rightarrow U \\ A \in \text{Ob } \mathbb{D}(V)}} v_!H_V(A, A), \quad (15)$$

and write $\iota_{v,A} : v_!H_V(A, A) \rightarrow P_U$ for its coproduct injections.

There are two sorts of relations. Fibrewise dinaturality is generated by

$$R_U^{\text{din}} = \coprod_{\substack{v: V \rightarrow U \\ f: A \rightarrow B \text{ in } \mathbb{D}(V)}} v! H_V(B, A). \quad (16)$$

On the summand indexed by $(v, f : A \rightarrow B)$, define the parallel maps to P_U by

$$d_0^{\text{din}} = \iota_{v,A} v! H_V(f^{\text{op}}, 1_A), \quad (17)$$

$$d_1^{\text{din}} = \iota_{v,B} v! H_V(1_B, f). \quad (18)$$

Base coherence is generated by

$$R_U^{\text{bas}} = \coprod_{\substack{v: V \rightarrow U, \ w: W \rightarrow V \\ A \in \text{Ob } \mathbb{D}(V)}} (vw)_! w^* H_V(A, A). \quad (19)$$

On the summand indexed by (v, w, A) , its two maps to P_U are

$$d_0^{\text{bas}} = (vw)_! w^* H_V(A, A) \xrightarrow{(vw)_! \chi_{w,A,A}^H} (vw)_! H_W(w^* A, w^* A) \xrightarrow{\iota_{vw, w^* A}} P_U, \quad (20)$$

$$d_1^{\text{bas}} = (vw)_! w^* H_V(A, A) \xrightarrow{\beta_{v,w, H_V(A,A)}} v! H_V(A, A) \xrightarrow{\iota_{v,A}} P_U. \quad (21)$$

Let

$$R_U = R_U^{\text{din}} \amalg R_U^{\text{bas}}, \quad d_i = d_i^{\text{din}} \amalg d_i^{\text{bas}} \quad (i = 0, 1). \quad (22)$$

Theorem 7.2 (Fixed-base coequalizer formula). *Suppose that every v^* has a chosen left adjoint $v_!$, and that $\mathbb{V}(U)$ has the coproducts in equations (15), (16) and (19), the binary coproduct in equation (22), and the coequalizer*

$$R_U \xrightarrow[d_1]{d_0} P_U \xrightarrow{q_U} L_U. \quad (23)$$

Then L_U , with the cowedge defined below, is the relative coend of H over U . In particular,

$$\mathbb{V}(U)(L_U, X) \cong \text{Cow}_U^{\text{rel}}(H; X)$$

naturally in $X \in \mathbb{V}(U)$.

No pullbacks in $\mathcal{C}$, Beck–Chevalley isomorphisms, or preservation of colimits by reindexing are needed for this fixed-base statement.

Proof. We give the construction and both directions of the representing bijection.

Step 1: the universal components. For each $v : V \rightarrow U$ and $A \in \mathbb{D}(V)$, transpose

$$v! H_V(A, A) \xrightarrow{\iota_{v,A}} P_U \xrightarrow{q_U} L_U$$

through $v_! \dashv v^*$. Denote the resulting map by

$$\lambda_{v,A}^U : H_V(A, A) \longrightarrow v^* L_U. \quad (24)$$

Step 2: fibrewise dinaturality. For $f : A \rightarrow B$ in $\mathbb{D}(V)$, the equation $q_U d_0^{\text{din}} = q_U d_1^{\text{din}}$ on the (v, f) -summand says

$$q_U \iota_{v,A} v! H_V(f^{\text{op}}, 1_A) = q_U \iota_{v,B} v! H_V(1_B, f).$$

Taking adjoints under $v_! \dashv v^*$ gives exactly

$$\lambda_{v,A}^U H_V(f^{\text{op}}, 1_A) = \lambda_{v,B}^U H_V(1_B, f).$$

Step 3: base coherence. On the (v, w, A) -summand of R_U^{bas} , the equation $q_U d_0^{\text{bas}} = q_U d_1^{\text{bas}}$ has two adjuncts under $(vw)_! \dashv (vw)^*$. The adjunct of the first is

$$w^* H_V(A, A) \xrightarrow{\chi_{w, A, A}^H} H_W(w^* A, w^* A) \xrightarrow{\lambda_{vw, w^* A}^U} (vw)^* L_U.$$

By [lemma 7.1](#), the adjunct of the second is

$$w^* H_V(A, A) \xrightarrow{w^* \lambda_{v, A}^U} w^* v^* L_U \xrightarrow{\alpha_{w, v, L_U}} (vw)^* L_U.$$

Their equality is precisely [equation \(4\)](#). Thus λ^U is a relative cowedge.

Step 4: maps out of the coequalizer give cowedges. Given $m : L_U \rightarrow X$, define

$$\omega_{v, A}^m = v^* m \lambda_{v, A}^U.$$

The equations already proved for λ^U , together with naturality of the pseudofunctor constraints, show that ω^m is a relative cowedge.

Step 5: every cowedge factors uniquely. Conversely, let $\omega \in \text{Cow}_U^{\text{rel}}(H; X)$. Transpose each component to

$$\bar{\omega}_{v, A} : v_! H_V(A, A) \longrightarrow X.$$

The coproduct property gives a unique $\bar{\omega} : P_U \rightarrow X$ with $\bar{\omega} \iota_{v, A} = \bar{\omega}_{v, A}$. On a summand of R_U^{din} , the equality $\bar{\omega} d_0 = \bar{\omega} d_1$ is, after adjunction, exactly the fibrewise dinaturality of ω . On a summand of R_U^{bas} , [lemma 7.1](#) identifies the same equality with the base-coherence equation for ω . Thus $\bar{\omega}$ coequalizes d_0, d_1 , so there is a unique $m : L_U \rightarrow X$ such that $m q_U = \bar{\omega}$. Taking adjuncts on each generator shows $v^* m \lambda_{v, A}^U = \omega_{v, A}$.

The two assignments are inverse because both are determined on the coproduct generators, and they are natural in X because postcomposition is used at every step. [Proposition 3.3](#) now identifies L_U as the relative coend. $\square$

Remark 7.3 (Relation to the ordinary formula). When $\mathcal{C} = \mathbf{1}$, the base-relation summands only encode unit constraints, and [equation \(23\)](#) reduces to the standard coequalizer by objects and arrows of $\mathbb{D}(\ast)$. This agrees with the ordinary definition as an initial cowedge [[Lor23](#), Definitions 1.1.4 and 1.1.6, pp. 23–24]. On a nonterminal base, the summand R_U^{bas} is additional data; deleting it changes the universal problem.

8 Base-change stability and the missing hypothesis

The fixed-base construction does not by itself imply that the maps $b_u : L_{U'} \rightarrow u^* L_U$ are invertible. The exact criterion is immediate from the universal property.

Proposition 8.1 (Exact stability criterion). *For $u : U' \rightarrow U$, assume that relative coends exist over U and U' . The canonical map b_u is an isomorphism if and only if the restricted universal cowedge*

$$\rho_u(L_U, \lambda^U) = (u^* L_U, \rho_u \lambda^U)$$

is initial in $\mathbf{Cow}_{U'}(H)$.

Proof. If b_u is invertible, it is an isomorphism in $\mathbf{Cow}_{U'}(H)$ from the initial cowedge $(L_{U'}, \lambda^{U'})$ to the restricted cowedge; hence the latter is initial. Conversely, if both cowedges are initial, there is a unique arrow in each direction between them. The two composites are endomorphisms of initial objects and therefore identities. The arrow from $(L_{U'}, \lambda^{U'})$ is, by its defining equation, b_u ; hence it is invertible. $\square$

The following example isolates a genuine gap in a tempting statement of the hypotheses.

Example 8.2 (Available Beck–Chevalley squares are not enough). Let $\mathcal{C}$ have three objects and two nonidentity arrows

$$0 \xrightarrow{p} 2 \xleftarrow{q} 1,$$

and equip it with the trivial Grothendieck topology. Let $\mathbb{V}$ be the constant indexed category **Set**, so every reindexing functor and every chosen left adjoint is the identity. All coproducts and coequalizers exist and are preserved, and the Beck–Chevalley map for every pullback square that exists is an identity.

Put

$$\mathbb{D}(2) = \emptyset, \quad \mathbb{D}(0) = \mathbb{D}(1) = \mathbf{1},$$

with the unique reindexing functors, and set $H_0(*, *) = H_1(*, *) = \mathbf{1}$. There is no component over 2.

A relative cowedge over 2 with apex X consists of one point of X contributed by $(p, *)$ and one independently contributed by $(q, *)$. There is no arrow in a slice relating these two components. Consequently

$$\text{Cow}_2^{\text{rel}}(H; X) \cong X \times X \quad \text{and} \quad L_2 \cong \{0, 1\}.$$

Over 0 and 1 there is one component, so $L_0 \cong L_1 \cong \mathbf{1}$. Since reindexing in $\mathbb{V}$ is the identity,

$$b_p : \mathbf{1} \longrightarrow \{0, 1\}, \quad b_q : \mathbf{1} \longrightarrow \{0, 1\}$$

are the two coproduct injections and are not isomorphisms.

Thus cocomplete fibres, cocontinuous reindexing, left adjoints, and Beck–Chevalley for all *available* pullback squares do not imply stability on an arbitrary site. What is missing here is the pullback of p and q . The next theorem shows that chosen pullbacks, together with the expected exactness, close precisely this gap for the displayed presentation.

Lemma 8.3 (Pasting for the canonical Beck–Chevalley mates). *For a chosen pullback square as in equation (27), let*

$$\text{BC}_{u,v} : s_! r^* \Longrightarrow u^* v_!$$

be the left mate of the pseudofunctorial comparison $r^ v^* \cong s^* u^*$. Given $w : T \rightarrow V$, form the pullback of vw along u ,*

$$Z \xrightarrow{k} T, \quad Z \xrightarrow{t} U',$$

and let $\bar{w} : Z \rightarrow W$ be determined by $r\bar{w} = wk$ and $s\bar{w} = t$. Then, with the canonical pseudofunctor constraints suppressed, use the following convention: if the induced square $(k, \bar{w}; r, w)$ is not the selected pullback of w along r , transport its canonical mate along the unique isomorphism to the selected pullback. Canonical mates and their invertibility are unchanged by this transport. With that convention,

$$\text{BC}_{u,vw} = u^*(c_{v,w}^{-1}) (\text{BC}_{u,v} w_!) s_! (\text{BC}_{r,w}) c_{s,\bar{w}}. \quad (25)$$

Moreover, for $Z_0 \in \mathbb{V}(V)$,

$$u^* \beta_{v,w,Z_0} \text{BC}_{u,vw,w^* Z_0} = \text{BC}_{u,v,Z_0} \beta_{s,\bar{w},r^* Z_0} t_! (\zeta_{Z_0}), \quad (26)$$

where $\zeta_{Z_0} : k^* w^* Z_0 \cong \bar{w}^* r^* Z_0$ is induced by $wk = r\bar{w}$.

Proof. Take right mates of both sides of equation (25), first under $t_! \dashv t^*$ and then under $(vw)_! \dashv (vw)^*$. The right mate of the left-hand side is, by definition, the comparison associated to the outer rectangle

$$k^* w^* v^* \cong t^* u^*.$$

On the right-hand side, the mates of $\text{BC}_{r,w}$ and $\text{BC}_{u,v}$ are the comparisons for the two constituent pullback squares, while the mates of $c_{s,\bar{w}}$ and $c_{v,w}^{-1}$ merely reassociate the two composite adjunctions. After the two triangle identities cancel the inserted units and counits, the result is the pasted comparison

$$k^*w^*v^* \cong \bar{w}^*r^*v^* \cong \bar{w}^*s^*u^* \cong t^*u^*.$$

The pseudofunctor pentagon says that this is the outer-rectangle comparison. Since taking mates is bijective, [equation \(25\)](#) follows.

For [equation \(26\)](#), expand each β by [equation \(14\)](#), substitute [equation \(25\)](#), and use naturality of $\text{BC}_{u,v}$ with respect to the counit $\varepsilon^w : w_!w^* \rightarrow 1$. The middle composite is then

$$s_! \bar{w}_! k^* w^* Z_0 \xrightarrow{s_! \bar{w}_! (\zeta_{Z_0})} s_! \bar{w}_! \bar{w}^* r^* Z_0 \xrightarrow{s_! \varepsilon^{\bar{w}}} s_! r^* Z_0,$$

on both sides. The remaining c 's agree by the triple coherence proved from uniqueness of composite adjoints above. This gives [equation \(26\)](#). The analogous replacements involving χ^H agree by the composition axiom for the indexed functor H ; that is the compatibility used on base-relation summands below. $\square$

Theorem 8.4 (Stable base change from pullbacks). *Assume the hypotheses of [theorem 7.2](#) for every base object. Assume in addition that:*

(S1) $\mathcal{C}$ has chosen pullbacks;

(S2) the canonical mates of the chosen adjoints satisfy Beck–Chevalley: for every chosen pullback square

$$\begin{array}{ccc} W & \xrightarrow{r} & V \\ s \downarrow & & \downarrow v \\ U' & \xrightarrow{u} & U \end{array} \quad \text{BC}_{u,v} : s_! r^* \xrightarrow{\cong} u^* v_!; \quad (27)$$

(S3) every u^* preserves the coproducts and coequalizers appearing in [equations \(15\)](#), [\(16\)](#), [\(19\)](#), [\(22\)](#) and [\(23\)](#), including the binary coproduct $R_U^{\text{din}} \amalg R_U^{\text{bas}}$.

Then every canonical base-change map

$$b_u : L_{U'} \xrightarrow{\cong} u^* L_U$$

is an isomorphism. Hence the fixed-base coequalizers assemble, uniquely, to a cartesian pseudonatural object of $\mathbb{V}$.

Proof. Fix $u : U' \rightarrow U$. We construct an inverse $a_u : u^* L_U \rightarrow L_{U'}$ on generators and then check both composites. Write

$$j_{v,A}^U = q_U \iota_{v,A} : v_! H_V(A, A) \longrightarrow L_U.$$

Step 1: pull a generator across the square. For every $v : V \rightarrow U$, choose its pullback along u as in [equation \(27\)](#). Preservation of the coproduct in [equation \(15\)](#), Beck–Chevalley, and the indexed comparison of H give

$$u^* P_U \cong \coprod_{v:V \rightarrow U, A \in \mathbb{D}(V)} u^* v_! H_V(A, A) \quad (28)$$

$$\xrightarrow{\amalg \text{BC}_{u,v}^{-1}} \coprod_{v,A} s_! r^* H_V(A, A) \quad (29)$$

$$\xrightarrow{\amalg s_! \chi_{r,A,A}^H} \coprod_{v,A} s_! H_W(r^* A, r^* A) \xrightarrow{\amalg \iota_{s,r^* A}} P_{U'} \xrightarrow{q_{U'}} L_{U'}. \quad (30)$$

Call this composite $a_P : u^*P_U \rightarrow L_{U'}$. Consequently, once a_u is induced from a_P , its defining generator equation is

$$a_u u^* j_{v,A}^{U'} = j_{s,r^*A}^{U'} s! \chi_{r,A,A}^H \text{BC}_{u,v,H_V(A,A)}^{-1}. \quad (31)$$

Step 2: the relations are respected. For a dinaturality summand indexed by $(v, f : A \rightarrow B)$, pullback turns f into $r^*f : r^*A \rightarrow r^*B$. Under the first three isomorphisms in [equation \(30\)](#), the two maps are exactly the two legs of the dinaturality relation indexed by (s, r^*f) in $R_{U'}^{\text{din}}$. They therefore agree after $q_{U'}$.

For a base-relation summand indexed by $v : V \rightarrow U$, $w : T \rightarrow V$, and $A \in \mathbb{D}(V)$, form the chosen pullbacks

$$W = U' \times_U V, \quad Z = U' \times_U T.$$

Writing $(s, r) : W \rightarrow U' \times V$ and $(t, k) : Z \rightarrow U' \times T$, the pullback universal property gives a unique $\bar{w} : Z \rightarrow W$ with

$$s\bar{w} = t, \quad r\bar{w} = wk.$$

Pasting of Beck–Chevalley squares identifies the pullback of the first leg [equation \(20\)](#) with the generator $(t, \bar{w}^*r^*A)$, and identifies the pullback of the second leg [equation \(21\)](#) with the generator (s, r^*A) . The comparison $\bar{w}^*r^*A \cong k^*w^*A$ is the pseudofunctorial constraint of $\mathbb{D}$; the analogous comparison for H is the coherence of χ^H . Precisely, [equation \(25\)](#) gives the comparison for the first leg, while [equation \(26\)](#) gives the comparison for the β -map in the second leg. The composition axiom for χ^H transports both to the same chosen identification of $\bar{w}^*r^*H_V(A, A)$ with $k^*w^*H_V(A, A)$. More explicitly, after these identifications the two composites into $L_{U'}$ are

$$\begin{aligned} t! \bar{w}^* H_W(r^*A, r^*A) &\xrightarrow{t! \chi_{\bar{w}^*r^*A, r^*A}^H} t! H_Z(\bar{w}^*r^*A, \bar{w}^*r^*A) \xrightarrow{q_{U'} \iota_{t, \bar{w}^*r^*A}} L_{U'}, \\ t! \bar{w}^* H_W(r^*A, r^*A) &\xrightarrow{\beta_{s, \bar{w}, H_W(r^*A, r^*A)}} s! H_W(r^*A, r^*A) \xrightarrow{q_{U'} \iota_{s, r^*A}} L_{U'}. \end{aligned}$$

They are precisely the two legs of the base relation in $R_{U'}^{\text{bas}}$ indexed by $(s, \bar{w}, r^*A)$. The second leg has the displayed direction by [lemma 7.1](#). Thus it too is equalized by $q_{U'}$.

It follows that a_P coequalizes u^*d_0, u^*d_1 . Since u^* preserves [equation \(23\)](#), there is a unique

$$a_u : u^*L_U \longrightarrow L_{U'} \quad \text{with} \quad a_u u^*q_U = a_P. \quad (32)$$

We shall also use the following generator form of the defining equation for b_u :

$$b_u j_{s,r^*A}^{U'} s! \chi_{r,A,A}^H = u^* j_{v,A}^U \text{BC}_{u,v,H_V(A,A)}. \quad (33)$$

Both sides have domain $s!r^*H_V(A, A)$. To verify the equality, take adjuncts under $s! \dashv s^*$. The left adjunct, using the defining identity $b_u \lambda^{U'} = \rho_u \lambda^U$, is the composite represented by

$$r^*H_V(A, A) \xrightarrow{\chi_{r,A,A}^H} H_W(r^*A, r^*A) \xrightarrow{\lambda_{vr,r^*A}^U} (vr)^*L_U \cong s^*u^*L_U.$$

The right adjunct is the same composite: this is the definition of the canonical mate $\text{BC}_{u,v}$, followed by the base-coherence equation for λ^U along r . Adjunction is bijective, proving [equation \(33\)](#).

Step 3: $a_u b_u = 1$. It suffices to precompose with every coproduct generator $v'_! H_{V'}(A', A') \rightarrow P_{U'} \rightarrow L_{U'}$, since a coequalizer is an epimorphism. Pull back the composite $uv' : V' \rightarrow U$ along u :

$$W = U' \times_U V' \xrightarrow{r} V', \quad W \xrightarrow{s} U'.$$

There is a canonical section $\delta = (v', 1_{V'}) : V' \rightarrow W$ with $s\delta = v'$ and $r\delta = 1_{V'}$. By the definition of b_u , the following section mate makes this statement explicit. Put $Z_0 = H_{V'}(A', A')$, and let

$$\hat{\alpha}_{\delta, r, Z_0} : \delta! Z_0 \longrightarrow r^* Z_0$$

be the adjunct of the inverse of the constraint $\delta^* r^* Z_0 \cong (r\delta)^* Z_0 \cong Z_0$. Define

$$\Gamma_{\delta, A'} : v'_! Z_0 \cong (s\delta)_! Z_0 \xrightarrow{c_{s, \delta}} s_! \delta_! Z_0 \xrightarrow{s_! \hat{\alpha}_{\delta, r, Z_0}} s_! r^* Z_0. \quad (34)$$

The base relation of $L_{U'}$ indexed by $(s, \delta, r^* A')$, together with $r\delta = 1$ and the composition coherence of χ^H , is exactly

$$j_{s, r^* A'}^{U'} s_! \chi_{r, A', A'}^H \Gamma_{\delta, A'} = j_{v', A'}^{U'}. \quad (35)$$

Precomposing equation (33), for the pullback of uv' along u , with $\Gamma_{\delta, A'}$, and then using equation (35), gives

$$b_u j_{v', A'}^{U'} = u^* j_{uv', A'}^U \text{BC}_{u, uv', Z_0} \Gamma_{\delta, A'}. \quad (36)$$

Now equations (31), (35) and (36) give the calculation

$$\begin{aligned} a_u b_u j_{v', A'}^{U'} &= a_u u^* j_{uv', A'}^U \text{BC}_{u, uv', Z_0} \Gamma_{\delta, A'} \\ &= j_{s, r^* A'}^{U'} s_! \chi_{r, A', A'}^H \Gamma_{\delta, A'} \\ &= j_{v', A'}^{U'}. \end{aligned}$$

The maps $j_{v', A'}^{U'}$ are jointly epimorphic because they are the coproduct injections followed by the coequalizer. Hence $a_u b_u = 1$.

Step 4: $b_u a_u = 1$. Because u^* preserves the coequalizer, $u^* q_U$ is an epimorphism. Together with preservation of equation (15), it is therefore enough to precompose with every $u^* j_{v, A}^U$. Equations equations (31) and (33) give, on the pullback square for u and v ,

$$\begin{aligned} b_u a_u u^* j_{v, A}^U &= b_u j_{s, r^* A}^{U'} s_! \chi_{r, A, A}^H \text{BC}_{u, v, H_V(A, A)}^{-1} \\ &= u^* j_{v, A}^U \text{BC}_{u, v, H_V(A, A)} \text{BC}_{u, v, H_V(A, A)}^{-1} \\ &= u^* j_{v, A}^U. \end{aligned}$$

Thus $b_u a_u = 1_{u^* L_U}$.

Thus $a_u = b_u^{-1}$. Coherent assembly now follows from theorem 4.3. $\square$

Remark 8.5 (Exact comparison with Shulman’s hypotheses). Shulman calls the existence of $f_! \dashv f^*$ together with Beck–Chevalley for pullback squares “indexed coproducts” [Shu13, Definition 2.4, p. 7]. His tensor-preservation condition is Definition 2.10, and indexed profunctor composition is the coequalizer of Lemma 3.25 [Shu13, pp. 8 and 21–22]. Theorem 8.4 is not a restatement of that lemma: our R_U^{bas} explicitly imposes relations between all generalized objects in $\mathcal{C}/U$, while Shulman’s formula composes enriched indexed profunctors. A comparison between the resulting objects still requires a typed identification of these two universal problems.

9 A Fubini theorem for stable relative coends

The ordinary Fubini theorem identifies a coend over a product category with either order of iterated coends [Lor23, Theorem 1.3.1, pp. 37–39]. In the relative setting an inner coend must itself vary as indexed data before an outer relative coend can even be typed. This is why stability, rather than mere fibrewise existence, is the correct hypothesis.

The indexed use of Fubini is not itself new: Pitts invokes it in the S -indexed setting in the proof of his product theorem [Pit85, proof of Theorem 2.3, p. 51]. More recently, Nickel reported joint work with Arkor on coends in compact closed virtual equipments, explicitly aiming to recover Fubini and to include $\text{Span}(\mathcal{E})$ as an example [NA26]. The available source for the latter is a seminar abstract, not a public manuscript. The result below addresses a different fixed-site question: when

a full-slice inner universal object is sufficiently stable to serve as the input of another full-slice coend.

Let

$$\mathbb{A}, \mathbb{B} : \mathcal{C}^{\text{op}} \longrightarrow \mathbf{Cat}$$

be small-fibred indexed categories, and let

$$H : \mathbb{A}^\vee \times_{\mathcal{C}} \mathbb{A} \times_{\mathcal{C}} \mathbb{B}^\vee \times_{\mathcal{C}} \mathbb{B} \longrightarrow \mathbb{V} \quad (37)$$

be an indexed functor. Put

$$\widehat{H}_V((A', B'), (A, B)) = H_V(A', A, B', B).$$

Then $\widehat{H}$ is an indexed functor on $(\mathbb{A} \times_{\mathcal{C}} \mathbb{B})^\vee \times_{\mathcal{C}} (\mathbb{A} \times_{\mathcal{C}} \mathbb{B})$.

Definition 9.1 (Simultaneous relative cowedge). For $U \in \mathcal{C}$ and $X \in \mathbb{V}(U)$, a simultaneous relative cowedge for H is a family

$$\omega_{v,A,B} : H_V(A, A, B, B) \longrightarrow v^* X$$

for $v : V \rightarrow U$, $A \in \mathbb{A}(V)$, and $B \in \mathbb{B}(V)$, which is dinatural separately in A and in B . Explicitly, for $f : A_0 \rightarrow A_1$ and $g : B_0 \rightarrow B_1$,

$$\omega_{v,A_0,B} H_V(f^{\text{op}}, 1_{A_0}, 1_B, 1_B) = \omega_{v,A_1,B} H_V(1_{A_1}, f, 1_B, 1_B), \quad (38)$$

$$\omega_{v,A,B_0} H_V(1_A, 1_A, g^{\text{op}}, 1_{B_0}) = \omega_{v,A,B_1} H_V(1_A, 1_A, 1_{B_1}, g). \quad (39)$$

The common domains are respectively $H_V(A_1, A_0, B, B)$ and $H_V(A, A, B_1, B_0)$. Finally, the family satisfies the joint base-coherence equation

$$\omega_{vw, w^* A, w^* B} \chi_{w,A,A,B,B}^H = \alpha_{w,v,X} w^* \omega_{v,A,B}.$$

Its set is denoted $\text{Cow}_U^{\text{rel,bi}}(H; X)$.

Lemma 9.2 (Product versus separate dinaturality). *There is a natural equality of cowedge sets*

$$\text{Cow}_U^{\text{rel}}(\widehat{H}; X) = \text{Cow}_U^{\text{rel,bi}}(H; X).$$

Proof. Dinaturality for a morphism $(f, g) : (A, B) \rightarrow (A', B')$ in a product fibre implies the two separate equations by taking g or f to be an identity. Conversely, factor

$$(f, g) = (1_{A'}, g)(f, 1_B).$$

Apply A -dinaturality to the first change of object and B -dinaturality to the second; functoriality of H_V makes the middle terms agree. This gives dinaturality for (f, g) . The base-coherence equation is literally the same for $\widehat{H}$, so the two sets coincide, naturally in X . $\square$

For $B', B \in \mathbb{B}(U)$, restrict H to the slice $\mathcal{C}/U$ by setting

$$H_w^{B',B}(A', A) = H_W(A', A, w^* B', w^* B) \quad (w : W \rightarrow U).$$

Assume that its relative coend exists and write

$$K_U(B', B) = \int_{\text{rel}, U}^{A \in \mathbb{A}} H(A, A, B', B). \quad (40)$$

Its universal components are

$$\kappa_{w,A}^{U,B',B} : H_W(A, A, w^* B', w^* B) \longrightarrow w^* K_U(B', B). \quad (41)$$

Whenever the inner coends over U and U' exist, write

$$b_u^{B',B} : K_{U'}(u^* B', u^* B) \longrightarrow u^* K_U(B', B) \quad (42)$$

for their canonical base-change map from [equation \(8\)](#).

Lemma 9.3 (The inner universal cowedge under base change). *For $u : U' \rightarrow U$, $w : W \rightarrow U'$, $A \in \mathbb{A}(W)$, and $B', B \in \mathbb{B}(U)$, the canonical map [equation \(42\)](#) is characterized by*

$$w^* b_u^{B', B} \kappa_{w, A}^{U', u^* B', u^* B} = \alpha_{w, u, K_U(B', B)}^{-1} \kappa_{uw, A}^{U, B', B} H_W(1, 1, (\alpha_{w, u, B'}^{-1})^{\text{op}}, \alpha_{w, u, B}). \quad (43)$$

The suppressed unit constraints are the same on both sides.

Proof. The right-hand side is the component at (w, A) of the restriction along u of the universal inner cowedge over U : the last arrow transports the parameters $w^* u^* B', w^* u^* B$ to $(uw)^* B', (uw)^* B$, and the preceding α^{-1} transports its apex to $w^* u^* K_U(B', B)$. By definition, $b_u^{B', B}$ is the unique map from the universal cowedge over U' to this restricted cowedge. Its component equation is therefore exactly [equation \(43\)](#). $\square$

Lemma 9.4 (Stable inner coends form an indexed bifunctor). *Suppose the objects in [equation \(40\)](#) exist for every U, B', B , and the canonical base-change maps in [equation \(42\)](#) are isomorphisms. Then the K_U assemble to an indexed functor*

$$K : \mathbb{B}^\vee \times_{\mathcal{C}} \mathbb{B} \longrightarrow \mathbb{V},$$

whose comparison along u is $(b_u^{B', B})^{-1}$.

Proof. First fix U . For arrows $r : B'_0 \rightarrow B'_1$ and $s : B_0 \rightarrow B_1$ in $\mathbb{B}(U)$, define

$$K_U(r^{\text{op}}, s) : K_U(B'_1, B_0) \longrightarrow K_U(B'_0, B_1)$$

as the unique map induced by the inner cowedge whose (w, A) -component is

$$H_W(A, A, w^* B'_1, w^* B_0) \xrightarrow{H_W(1, 1, (w^* r)^{\text{op}}, w^* s)} H_W(A, A, w^* B'_0, w^* B_1) \xrightarrow{\kappa_{w, A}^{U, B'_0, B_1}} w^* K_U(B'_0, B_1).$$

For $z : Z \rightarrow W$, the base-coherence equation for this candidate is the base-coherence equation for κ_{U, B'_0, B_1}^U , pasted with the indexed naturality square

$$\chi_z^H z^* H_W(1, 1, (w^* r)^{\text{op}}, w^* s) = H_Z(1, 1, ((wz)^* r)^{\text{op}}, (wz)^* s) \chi_z^H,$$

where the parameter α 's are suppressed. Thus indexed naturality of H , not fibrewise naturality alone, shows that this is an inner relative cowedge. Two successive such maps and the map associated to the composite have identical composites with every universal component $\kappa_{w, A}$, so uniqueness in the inner universal property makes them equal. The same argument with identity arrows proves the identity law. Thus each K_U is a bifunctor with the asserted variance.

For $u : U' \rightarrow U$, set

$$\chi_{u, B', B}^K = (b_u^{B', B})^{-1} : u^* K_U(B', B) \xrightarrow{\cong} K_{U'}(u^* B', u^* B).$$

For $r : B'_0 \rightarrow B'_1$ and $s : B_0 \rightarrow B_1$, naturality of the base map is the equality

$$u^* K_U(r^{\text{op}}, s) b_u^{B'_1, B_0} = b_u^{B'_0, B_1} K_{U'}((u^* r)^{\text{op}}, u^* s). \quad (44)$$

Precompose both sides with every $\kappa_{w, A}^{U'}$ and apply [equation \(43\)](#). After cancelling the same α 's, the two resulting maps differ only by whether the indexed naturality square of H is traversed

before or after $H(1, 1, (w^*u^*r)^{\text{op}}, w^*u^*s)$; hence that square proves [equation \(44\)](#). Universality of the inner coend then gives the displayed equality.

For composable $U'' \xrightarrow{v} U' \xrightarrow{u} U$, put

$$\begin{aligned} a' &= \alpha_{v,u,B'}^{\mathbb{B}} : v^*u^*B' \longrightarrow (uv)^*B', \\ a &= \alpha_{v,u,B}^{\mathbb{B}} : v^*u^*B \longrightarrow (uv)^*B, \end{aligned}$$

and define the parameter transport

$$\Theta_{v,u}^{B',B} = K_{U''}((a')^{\text{op}}, a^{-1}) : K_{U''}((uv)^*B', (uv)^*B) \longrightarrow K_{U''}(v^*u^*B', v^*u^*B). \quad (45)$$

The correctly typed composition equation for the base maps is

$$b_{uv}^{B',B} = \alpha_{v,u,K_U(B',B)}^{\mathbb{V}} v^*b_u^{B',B} b_v^{u^*B',u^*B} \Theta_{v,u}^{B',B}. \quad (46)$$

Indeed, precompose both sides with every universal component for $K_{U''}((uv)^*B', (uv)^*B)$. Applying [equation \(43\)](#) once to the left and successively for v and u to the right reduces both sides to the same component of $\kappa^{U,B',B}$. The two parameter transports agree by the pentagon for $\mathbb{B}$, the two apex transports agree by the pentagon for $\mathbb{V}$, and the intervening maps agree by the composition axiom for χ^H . Universality of the inner coend proves [equation \(46\)](#).

For completeness, if $\iota_{U,B}^{\mathbb{B}} : 1_U^*B \rightarrow B$ and $\iota_{U,X}^{\mathbb{V}} : 1_U^*X \rightarrow X$ are the unitors, the corresponding typed unit equation is

$$\iota_{U,K_U(B',B)}^{\mathbb{V}} b_{1_U}^{B',B} K_U((\iota_{U,B'}^{\mathbb{B}})^{\text{op}}, (\iota_{U,B}^{\mathbb{B}})^{-1}) = 1_{K_U(B',B)}. \quad (47)$$

It follows by the same one-component calculation, using the triangle axioms instead of the pentagons. Inverting [equation \(46\)](#) gives the explicit indexed-functor composition law

$$\chi_{v,u^*B',u^*B}^K v^* \chi_{u,B',B}^K = \Theta_{v,u}^{B',B} \chi_{uv,B',B}^K \alpha_{v,u,K_U(B',B)}^{\mathbb{V}}. \quad (48)$$

[Equation \(47\)](#) gives its unit law, and [equation \(44\)](#) gives its naturality. Therefore K is an indexed functor. $\square$

Proposition 9.5 (Currying simultaneous cowedges). *Under the hypotheses of [lemma 9.4](#), there is a natural bijection*

$$\text{Cow}_U^{\text{rel,bi}}(H; X) \cong \text{Cow}_U^{\text{rel}}(K; X) \quad (49)$$

for every U and $X \in \mathbb{V}(U)$.

Proof. We construct both maps and verify every axiom.

From a simultaneous cowedge to an outer cowedge. Let ω be simultaneous. Fix $v : V \rightarrow U$ and $B \in \mathbb{B}(V)$. For $w : W \rightarrow V$ and $A \in \mathbb{A}(W)$, define

$$H_W(A, A, w^*B, w^*B) \xrightarrow{\omega_{vw,A,w^*B}} (vw)^*X \xrightarrow{\alpha_{w,v,X}^{-1}} w^*v^*X. \quad (50)$$

The A -dinaturality of ω proves fibrewise dinaturality of this family. For $z : Z \rightarrow W$, its base-coherence square is the joint base-coherence square of ω , pasted with the associativity pentagon for α . Hence [equation \(50\)](#) is an inner relative cowedge over V with apex v^*X . Universality of $K_V(B, B)$ supplies a unique map

$$\bar{\omega}_{v,B} : K_V(B, B) \longrightarrow v^*X \quad (51)$$

whose pullback along every w , after $\kappa_{w,A}^{V,B,B}$, is [equation \(50\)](#).

We check B -dinaturality. For $f : B \rightarrow C$ in $\mathbb{B}(V)$, the two required maps from $K_V(C, B)$ to v^*X are

$$\bar{\omega}_{v,B} K_V(f^{\text{op}}, 1_B), \quad \bar{\omega}_{v,C} K_V(1_C, f).$$

Precompose their pullbacks with every universal component of $K_V(C, B)$. By the definition of the two actions of K , the resulting maps are respectively

$$\begin{aligned} H_W(A, A, w^*C, w^*B) &\xrightarrow{H(1,1,(w^*f)^{\text{op}},1)} H_W(A, A, w^*B, w^*B) \longrightarrow w^*v^*X, \\ H_W(A, A, w^*C, w^*B) &\xrightarrow{H(1,1,1,w^*f)} H_W(A, A, w^*C, w^*C) \longrightarrow w^*v^*X. \end{aligned}$$

They agree by the B -dinaturality of ω . The representing property of the inner coend then makes the two maps from $K_V(C, B)$ equal.

For base coherence, let $w : W \rightarrow V$. We must show

$$\bar{\omega}_{vw,w^*B} \chi_{w,B,B}^K = \alpha_{w,v,X} w^* \bar{\omega}_{v,B} : w^* K_V(B, B) \longrightarrow (vw)^* X. \quad (52)$$

Precompose both sides with $b_w^{B,B} : K_W(w^*B, w^*B) \rightarrow w^* K_V(B, B)$, which is an isomorphism. Then precompose with every $\kappa_{z,A}^{W,w^*B,w^*B}$. On the left, $\chi_{w,B,B}^K b_w^{B,B} = 1$, so the defining property of $\bar{\omega}_{vw,w^*B}$ gives

$$H_Z(A, A, z^*w^*B, z^*w^*B) \xrightarrow{\omega(vw)z,A,z^*w^*B} ((vw)z)^* X \xrightarrow{\alpha_{z,vw,X}^{-1}} z^*(vw)^* X,$$

with the parameter constraints understood. On the right, [equation \(43\)](#) converts $\kappa_{z,A}^W$ followed by z^*b_w into the (wz, A) -component of κ^V . The defining property of $\bar{\omega}_{v,B}$ then gives the same displayed map; the equality of the two structural composites is the pentagon for α . Universality of $K_W(w^*B, w^*B)$, followed by invertibility of b_w , proves [equation \(52\)](#). Hence $\bar{\omega}$ is an outer relative cowedge for K .

From an outer cowedge to a simultaneous cowedge. Conversely, let $\tau \in \text{Cow}_U^{\text{rel}}(K; X)$. Define

$$\omega_{v,A,B}^\tau : H_V(A, A, B, B) \xrightarrow{\kappa_{1V,A}^{V,B,B}} K_V(B, B) \xrightarrow{\tau_{v,B}} v^* X, \quad (53)$$

with the unit constraints suppressed. The inner cowedge equation for κ gives A -dinaturality. The definition of the two actions of K , followed by the B -dinaturality of τ , gives B -dinaturality. For $w : W \rightarrow V$, the required compatibility is not implicit: paste the inner base-coherence equation for the universal cowedge κ with [equation \(43\)](#) for $u = w$ and the identity arrow of W . Postcomposing that equality with the outer base-coherence equation for τ gives

$$\omega_{vw,w^*A,w^*B}^\tau \chi_{w,A,A,B,B}^H = \alpha_{w,v,X} w^* \omega_{v,A,B}^\tau.$$

Thus ω^τ is simultaneous.

The constructions are inverse. Starting with ω , the uncurried component at (v, A, B) evaluates the map $\bar{\omega}_{v,B}$ on the identity-base universal component. Its defining property [equation \(50\)](#) therefore returns $\omega_{v,A,B}$, up to the unit constraint. Starting with τ , the inner cowedge used to construct $\bar{\omega}_{v,B}$ from ω^τ is the restriction of $\tau_{v,B}$ along all $w : W \rightarrow V$. More explicitly, [equation \(43\)](#), together with the base-coherence of the inner universal cowedge and the unit constraints, gives

$$\kappa_{1W,A}^{W,w^*B,w^*B} = \chi_{w,B,B}^K \kappa_{w,A}^{V,B,B}. \quad (54)$$

The outer base-coherence axiom for τ then yields

$$\alpha_{w,v,X}^{-1} \tau_{vw,w^*B} \kappa_{1W,A}^{W,w^*B,w^*B} = w^* \tau_{v,B} \kappa_{w,A}^{V,B,B}. \quad (55)$$

Thus this inner cowedge is exactly the restriction of $\tau_{v,B}$. The uniqueness clause for the inner coend therefore gives $\bar{\omega}_{v,B} = \tau_{v,B}$. Both assignments commute with postcomposition in X , proving naturality. $\square$

Theorem 9.6 (Relative Fubini). *Assume that the $\mathbb{A}$ -inner coends $K_U(B', B)$ of [equation \(40\)](#) exist and are stable under base change. Then the simultaneous and iterated representing objects below have equivalent existence: either both exist or neither does. When they exist, there is a canonical isomorphism*

$$\int_{\text{rel}, U}^{(A, B) \in \mathbb{A} \times \mathbb{B}} H(A, A, B, B) \cong \int_{\text{rel}, U}^{B \in \mathbb{B}} K(B, B). \quad (56)$$

Here the right-hand side is the relative coend of the indexed bifunctor

$$K_V(B', B) = \int_{\text{rel}, V}^{A \in \mathbb{A}} H(A, A, B', B),$$

not a list of unrelated fibrewise coends.

If, symmetrically, the $\mathbb{B}$ -inner coends exist stably, then whenever one of the simultaneous or iterated coends exists, all corresponding two-variable admissible orders exist and are canonically isomorphic.

Proof. By [lemma 9.2](#) and [proposition 9.5](#), for every $X \in \mathbb{V}(U)$ there are natural bijections

$$\text{Cow}_U^{\text{rel}}(\widehat{H}; X) = \text{Cow}_U^{\text{rel}, \text{bi}}(H; X) \cong \text{Cow}_U^{\text{rel}}(K; X).$$

The simultaneous coend is, by [proposition 3.3](#), a representing object for the first functor of X ; the iterated coend is a representing object for the last. The natural bijection therefore transports a universal element in either direction. Hence one object exists if and only if the other does, and uniqueness of representing objects gives the canonical isomorphism [equation \(56\)](#).

Repeating the proof with $\mathbb{A}$ and $\mathbb{B}$ interchanged gives the other iterated order. Every comparison is the unique isomorphism that transports the same simultaneous universal cowedge. No higher-variable coherence is claimed here: it requires a separately defined multivariable relative cowedge and recursive stability hypotheses for each intermediate coend. $\square$

Remark 9.7 (Why stability is not cosmetic). If the inner objects $K_U(B', B)$ merely exist separately, the canonical maps in [equation \(42\)](#) need not be invertible. Then they do not provide the indexed comparison in the direction required for an indexed functor K , so the outer relative coend in [equation \(56\)](#) is not typed by the present definition. The theorem therefore does not silently promote fibrewise existence to indexed Fubini. Under the hypotheses of [theorem 8.4](#), the needed stability follows automatically from the coequalizer presentation.

10 Further directions and present limits

Theorems [7.2](#), [8.4](#), and [9.6](#) establish fixed-base existence, one checkable stable range, and Fubini. They do not yet identify the relative coend with Shulman's canceling tensor, prove the relative co-Yoneda formula announced in Drioux's slides, construct a profunctor equipment, or remove dependence on a chosen site presentation. The remaining problems are the following.

Problem 10.1 (Comparison with indexed canceling products). When Shulman's hypotheses hold, determine whether the localized relative coend is represented by his canceling tensor product. State separately the comparison on a fixed site and the additional descent needed for a stack of coefficients.

Problem 10.2 (Relative co-Yoneda). For the indexed hom object suggested by [equation \(13\)](#), identify the coefficient system and tensor needed to type the expression

$$\int_{\text{rel}}^A \mathbb{D}(A, X) \otimes F(A),$$

then prove a stable equivalence with $F(X)$. The proof must distinguish presheaf-level computation, sheafification, and descent.

Problem 10.3 (Relative profunctors). Use the existence and Fubini theorems to define composition of localization-sensitive indexed profunctors. Prove associativity and units, and compare the result with Shulman’s indexed profunctor equipment and Lee’s 2-categorical construction [\[Lee23\]](#).

Problem 10.4 (Site-presentation invariance). Let two small or small-generated sites present equivalent base topoi. Find the additional comparison data on $\mathbb{D}, \mathbb{V}, H$ for which their relative-coend objects correspond. A satisfactory theorem should be invariant under the relevant site Morita equivalence, not merely under a fibrewise equivalence chosen on one presentation.

These boundaries separate the present fixed-site calculus from the stronger relative-topos and Morita-invariant theory that remains to be developed.

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